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REPORT

Malware Sample Handling and Preliminary Static Analysis Report

Introduction

This report documents the preparation, acquisition, verification, and preliminary static inspection of a malware-analysis sample in an isolated Kali Linux virtual machine. The objective was to establish a safe analysis environment, preserve the original sample, identify the structure of the supplied archive, and prepare the sample for the approved static-analysis and UPX-packing workflow.

The analysis was conducted without executing the protected sample. Because the supplied sample is password-protected and the repository did not identify an authorized password for this particular file, the analysis could not proceed to extraction, hash calculation of the executable, UPX inspection, or unpacking.

Analysis Environment and Safety

The analysis was performed in a Kali Linux virtual machine running under VMware.

The virtual machine was configured for an isolated laboratory environment. Host-only networking was used rather than exposing the analysis VM directly to an external network.

The following safety principles were observed:

- The sample was handled inside the isolated analysis environment.

- The original downloaded archive was preserved.

- The protected sample was not executed.

No attempt was made to bypass or crack the ZIP password.

Password files belonging to unrelated laboratory samples were not used.

The sample was not uploaded to public malware-analysis services.

The original archive was retained for evidence preservation.

These precautions reduce the possibility of unintentionally exposing the sample or allowing it to interact with external systems.

Sample Acquisition

The original repository archive was located at:

`/home/obi/Downloads/malwares-main.zip`

Inspection of the archive confirmed that it contains several laboratory resources and malware-analysis samples.

The relevant sample was identified as:

`malwares-main/malware-sample-3 (1).zip`

The archive was confirmed to be a separate file in the root of the repository rather than a file contained inside one of the numbered laboratory directories.

Repository Structure Examination

The repository contents were examined to determine whether an accompanying password file existed for the selected sample.

Several `password.txt` files were found in the repository. However, these files were associated with specific laboratory samples, including:

1-1.BasicStaticAnalysis/Malware.PackedAndNotPacked.exe.malz/password.txt

1-2.DynamicAnalysis/Malware.TrojanSample.exe.malz/password.txt

2-1.ReverseEngineering/Malware.RootkitSample.exe.malz/password.txt

The selected sample,

`malware-sample-3 (1).zip`

was not located beside any of these password files.

The repository README was also searched for references to `password.txt`. The README documents password files associated with the individual laboratory directories, but

it does not provide a password or password convention specifically associated with `malware-sample-3 (1).zip`.

Therefore, the passwords associated with the other laboratory samples were not assumed to apply to this sample.

Archive Inspection

The selected archive was `mimalware-sample-3/`

1. `malware-sample-3/sample.bin`

The directory entry was not encrypted, while `sample.bin` was identified as encrypted/password-protected.

The archive metadata showed:

Archive size: approximately 779,368 bytes

Encrypted file: `malware-sample-3/sample.bin`

Compressed size: approximately 779,034 bytes

Reported uncompressed size: 1,217,024 bytes

Compression method: Deflated

CRC-32: 25325a46

The presence of the encryption flag prevented the protected file from being extracted without the correct password.

Password Investigation

A targeted investigation was performed to determine whether the repository contained an authorized password for the selected sample.

The following approaches were used:

Examination of the repository ZIP structure.

Identification of all relevant `password.txt` files.

Examination of the README around references to `password.txt`.

Searching for references to the sample name.

Checking whether a password.txt existed directly beside malware-sample-3 (1).zip.

The investigation established that the available password.txt files belong to other laboratory samples.

The README also did not identify a password for malware-sample-3 (1).zip.

Consequently, no password was guessed, reused from another sample, or cracked.

Current Analysis Status

The following activities have been completed:

Activity	Status
Kali Linux analysis VM prepared	Completed
Isolated networking configuration	Completed
Original repository archive identified	Completed
Target sample identified	Completed
Repository structure examined	Completed
Password files investigated	Completed
README password references examined	Completed
Target ZIP metadata inspected	Completed
Protected sample.bin identified	Completed
Sample executable extracted	Not completed
SHA-256 of sample.bin	Pending extraction
File-type identification of sample.bin	Pending extraction
UPX packing inspection	Pending extraction
Original/unpacked comparison	Pending
Approved UPX unpacking	Pending
Runtime analysis	Not performed

Network behavior analysis

Not performed

Limitations

The principal limitation was the password protection applied to `sample.bin`.

Because the correct authorized password was not identified in the repository documentation, extraction could not be completed. As a result, the executable itself could not yet be subjected to static inspection.

This means that no conclusion should currently be made about:

- Whether the executable is UPX-packed.

- Its PE sections.

- Its import table.

- Its entropy.

- Its SHA-256 hash.

- Its runtime behavior.

- Its registry or file-system activity.

- Its network communications.

- Any anti-analysis techniques.

These areas remain pending until the authorized password is obtained and the executable can be safely extracted.

Evidence Preservation

The original repository archive remains preserved at:

`/home/obi/Downloads/malwares-main.zip`

The original archive should be treated as the evidence source and should not be modified.

Any future extracted or unpacked version should be saved under a new filename or directory. Its SHA-256 should be calculated separately so that the original and derived files can be distinguished.

This approach maintains evidence integrity and provides a clear chain between the original sample and any analysis copies.

Preliminary Findings

The investigation produced the following findings:

Finding 1 Protected sample

The selected ZIP contains an encrypted file named `malware-sample-3/sample.bin`. The file cannot be extracted without an authorized password.

Finding 2 No sample-specific password identified

The repository contains multiple `password.txt` files, but they are associated with individual laboratory samples. No password file was identified specifically for `malware-sample-3 (1).zip`.

Finding 3 README does not resolve the password

The README references password files used by laboratory samples but does not identify an authorized password for the selected root-level sample.

Finding 4 No password bypass performed

No password cracking or bypass technique was attempted. This preserves the controlled nature of the laboratory exercise and avoids altering the evidence-handling procedure.

Conclusion

The preliminary malware-analysis workflow successfully established the location and structure of the supplied sample while maintaining evidence-handling precautions.

The original repository archive was identified and preserved. The target archive, `malware-sample-3 (1).zip`, was located and inspected. Its contents were shown to include a password-protected `sample.bin`. Repository documentation and available password files were examined, but none was confirmed to provide the authorized password for this particular sample.

The analysis therefore remains at the **pre-extraction stage**.

The next authorized step is to obtain the correct password from the course or instructor materials. Once the password is available, the protected executable can be extracted without modifying the original archive, its SHA-256 can be recorded, and the approved static-analysis workflow can continue with file-type identification, UPX detection, section/import/entropy comparison, and—if confirmed and authorized—UPX unpacking.

No claims about the executable's actual behavior or malicious characteristics are made at this stage because the executable itself has not yet been extracted or executed.

N/B lost screenshots for this lab

